Finite W-Algebras via BRST Cohomology

Ref 1 Arakawa, Arxiv 1605.00138

Ref 2 Boer, Tjin, Arxiv 9211109

General Notations

- $\mathfrak{g} \to$ f.d. (semi) simple Lie algebra, $(\cdot|\cdot) \to$ killing form $(x|y) = \mathrm{tr}(xy)$ or $\mathrm{tr}(\mathrm{ad}_x \mathrm{ad}_y)$
- $\{e, f, h\} \to \mathfrak{sl}_2$ - triple
- $S_f = f + \mathfrak{g}^e$ where $\mathfrak{g}^e = \{x \in \mathfrak{g} : [e, x] = 0\}$
- $\mathfrak{g}_j = \{x \in \mathfrak{g} : [h, x] = jx\}$ so that $\mathfrak{g} = \bigoplus_{j \in \mathbb{Z}} \mathfrak{g}_j$
- $\mathfrak{g}_1$ admits a symplectic form $\omega(x, y) = (f|[x, y])$

  Choose a Lagrangian subspace $\mathfrak{l} \subset \mathfrak{g}_1$ w.r.t. ω
- $\mathfrak{m} = \mathfrak{l} + \sum_{j \geq 2} \mathfrak{g}_j$. Then $\mathfrak{m}$ is a nilpotent subalgebra of $\mathfrak{g}$.
- $\chi : \mathfrak{m} \longrightarrow \mathbb{C}$ given by $\chi(x) = (f|x)$ defines a character.

  that is, $\chi([\mathfrak{m}, \mathfrak{m}]) = 0$
- M be unipotent group s.t. $\mathrm{Lie}(M) = \mathfrak{m}$.
- Adjoint action of M on $\mathfrak{g}$ is Hamiltonian.

  $\mu : \mathfrak{g}^* \to \mathfrak{m}^*$ (the restriction map) is moment map of adjoint action.
- $S_f \cong \mu^{-1}(\chi)/M$. (Slodowy Slice)

- For $\mathfrak{g} = \mathfrak{sl}_n$, m can be realized as positive eigenspace of $\mathrm{ad}\, \xi$ for some $\xi \in \mathfrak{g}$.

Let $\lambda = (\lambda_1, \lambda_2, \dots)$ be a partition of n.

Let $s_t = \sum_{k=1}^{t} \lambda'_k$ where λ' is the conjugate partition, namely $\lambda'_i = \#k$ for which $\lambda_k \geq i$

Lemma

Let $t_+ = \sum_{l \geq 1} \sum_{k=1}^{\lambda_{l-1}} E_{l+s_{k-1}, l+s_k}$,

(E: matrix units)

$$t_- = \sum_{l \geq 1} \sum_{k=1}^{\lambda_{l-1}} k(\lambda_l - k) E_{l+s_k, l+s_{k-1}} ,$$

$$t_0 = \sum_{l \geq 1} \sum_{k=1}^{\lambda_l} (\lambda_l + 1 - 2k) E_{l+s_{k-1}, l+s_{k-1}} .$$

Then $\{t_+, t_-, t_0\}$ form an $\mathfrak{sl}_2$-triple ∎

(*) t_0 is block diagonal with i^{th} block the size of $\lambda'_i \times \lambda'_i$.

- Let $\delta = \sum_{l \geq 1} \sum_{k=1}^{\lambda'_l} \left(\frac{\sum_j j \lambda'_j}{\sum_j \lambda'_j} - l \right) E_{s_{l-1}+k, s_{l-1}+k}$

Let $\mathfrak{g} = \bigoplus_j \mathfrak{g}_j$ where $\mathfrak{g}_j$ is j-eigenspace of $\operatorname{ad} \xi$.

Lemma

$\mathfrak{m}$ corresponding to the $\mathfrak{sl}_2$-triple $\{t_\pm, t_0\} \in \mathfrak{sl}_n$ is given by $\mathfrak{g}_+ = \bigoplus_{j>0} \mathfrak{g}_j$.

Correspondingly let $\mathfrak{g}_- = \bigoplus_{j<0} \mathfrak{g}_j$.

Then $\mathfrak{g} = \mathfrak{g}_- \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_+$.

Example

Let $\mathfrak{g} = \mathfrak{sl}_3$ and $\lambda = (2, 1)$.

Then $t_+ = E_{13}$, $t_- = E_{31}$, $t_0 = E_{11} - E_{33}$.

$\xi = \frac{1}{3}(E_{11} + E_{22} - 2E_{33})$

$\mathfrak{m} = \langle \{E_{13}, E_{23}\} \rangle = \mathfrak{g}_+$

$\chi(E_{13}) = 1, \quad \chi(E_{23}) = 0$

$\Rightarrow \mathfrak{g}_- = \langle \{E_{31}, E_{32}\} \rangle$.

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

grading w.r.t. $\operatorname{ad} \xi$.

The BRST Complex

A Clifford algebra

Let $Cl = Cl(\mathfrak{g}_+ \oplus \mathfrak{g}_+^*)$ be the Clifford algebra of the vector space $\mathfrak{g}_+ \oplus \mathfrak{g}_+^*$ w.r.t. the form $Q(x_1 + y_1^*, x_2 + y_2^*) = y_1^*(x_2) + y_2^*(x_1)$.

Then $Cl \cong \Lambda(\mathfrak{g}_+) \otimes \Lambda(\mathfrak{g}_+^*)$ as vector spaces, $\Lambda(\mathfrak{g}_+) \hookrightarrow Cl$ and $\Lambda(\mathfrak{g}_-) \hookrightarrow Cl$ are homogeneous embeddings of superalgebras, and $x y^* + y^* x = y^*(x)$.

(*) Latin indices run over a basis of $\mathfrak{g}$, Greek indices run over a basis of $\mathfrak{g}_+$, barred Greek indices run over a basis of $\mathfrak{g}_- \oplus \mathfrak{g}_0$.

- Choose a basis x_a of $\mathfrak{g}$ and let f_{ab}^c be structure constants: $[x_a, x_b] = \sum_c f_{ab}^c x_c$.
- For $x_\alpha \in \mathfrak{g}_+$, let $\xi_\alpha, \xi_\alpha^* \in Cl$ be the elements in Cl, corresponding to x_α and x_α^*.

The complex

- Let $\Omega = U(\mathfrak{f}) \otimes Cl$.

- Define a $\mathbb{Z}$-grading on Ω by $\deg(x_\alpha \otimes 1) = 0$,

$$\deg(1 \otimes \xi_\alpha^*) = +1, \quad \deg(1 \otimes \xi_\alpha) = -1.$$

Then $\Omega = \bigoplus_k \Omega_k$ with Ω_k being space of degree k elements

- Define a super-bracket on Ω by:

$$[A, B] = AB - (-1)^{\deg(A)\deg(B)} BA.$$

The differential

Let $Q = \sum_\alpha (x_\alpha - \mathcal{X}(x_\alpha)) \otimes \xi_\alpha^* - \frac{1}{2} \otimes \sum_{\alpha,\beta,\gamma} f_{\alpha\beta}^\gamma \xi_\gamma \xi_\alpha^* \xi_\beta^* \in \Omega$.
(BRST charge)

- Note that $\deg(Q) = 1$.

Lemma

Let $d(x) = [Q, x]$, for $x \in \Omega$.

Then $d^2 = 0$. ▨

(*) We now want to compute the cohomology

$$H^*(\Omega; d) := \ker d / \operatorname{Im} d.$$

The double complex

Write Q as $Q = Q_0 + Q_1$ where

$$Q_0 = \sum_\alpha x_\alpha \otimes \xi_\alpha^* - \frac{1}{2} \otimes \sum_{\alpha,\beta,\gamma} f_{\alpha\beta}^\gamma \, \xi_\gamma \, \xi_\alpha^* \, \xi_\beta^* \,,$$

$$Q_1 = \sum_\alpha - \chi(x_\alpha) \otimes \xi_\alpha^* \,.$$

Let $d_0(X) = [Q_0, X]$ and $d_1(X) = [Q_1, X]$, $X \in \Omega$.

Lemma

$$d_0^2 = d_0 d_1 + d_1 d_0 = d_1^2 = 0$$

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- Define a $(\mathbb{Z}\times\mathbb{Z})$-bigrading on Ω by:

$$\deg(x_a \otimes 1) = (k, -k) \text{ if } x_a \in \mathcal{A}_k,$$
$$\deg(1 \otimes \xi_\alpha^*) = (1-k, k) \text{ if } x_\alpha \in \mathcal{A}_k,$$
$$\deg(1 \otimes \xi_\alpha) = (k-1, -k) \text{ if } x_\alpha \in \mathcal{A}_k.$$

Lemma

The action of d_0 and d_1 is given explicity by:

$$d_0(x_a \otimes 1) = \sum_{\alpha,b} f_{\alpha a}^b (x_b \otimes \xi_\alpha^*), \quad d_1(x_a \otimes 1) = 0,$$

$$d_0(1 \otimes \xi_\alpha^*) = -\frac{1}{2} \sum_{\beta,\gamma} f_{\beta\gamma}^\alpha \, \xi_\beta^* \, \xi_\gamma^*, \quad d_1(1 \otimes \xi_\alpha^*) = 0,$$

$$d_0(1 \otimes \xi_\alpha) = (x_\alpha \otimes 1) + \sum_{\beta,\gamma} f_{\alpha\beta}^\gamma \, \xi_\gamma \, \xi_\beta^*, \quad d_1(1 \otimes \xi_\alpha) = -\chi(x_\alpha).$$

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